# Week 10.11 - 14.11

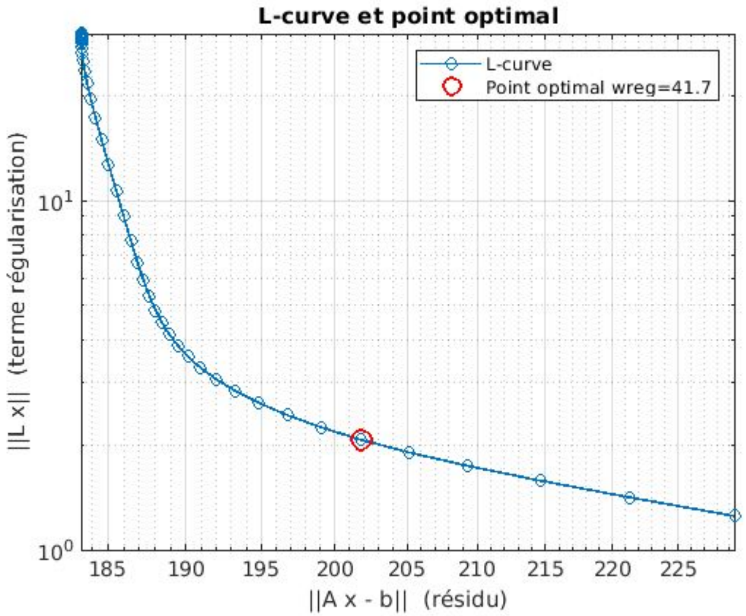
coding:

- Error field matlab code ->  L-curve for weight
- try different displacement in errorbar range
- compute manifold for more shots

- ridge regression, L-curve for tuning weight:

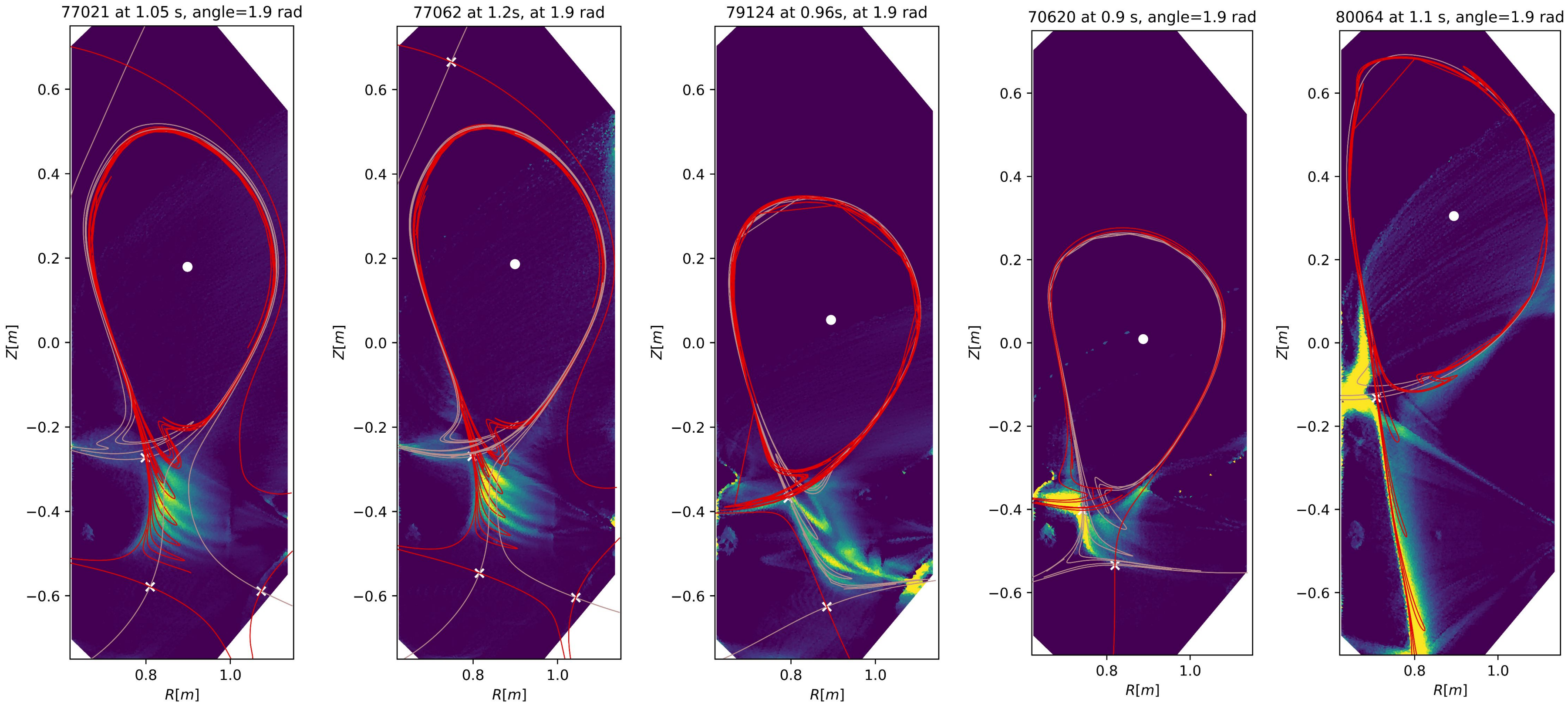
$$min||Ax - b||^2 + w_{reg}||x||^2$$

- approximation in Greenem, source of inaccuracy?

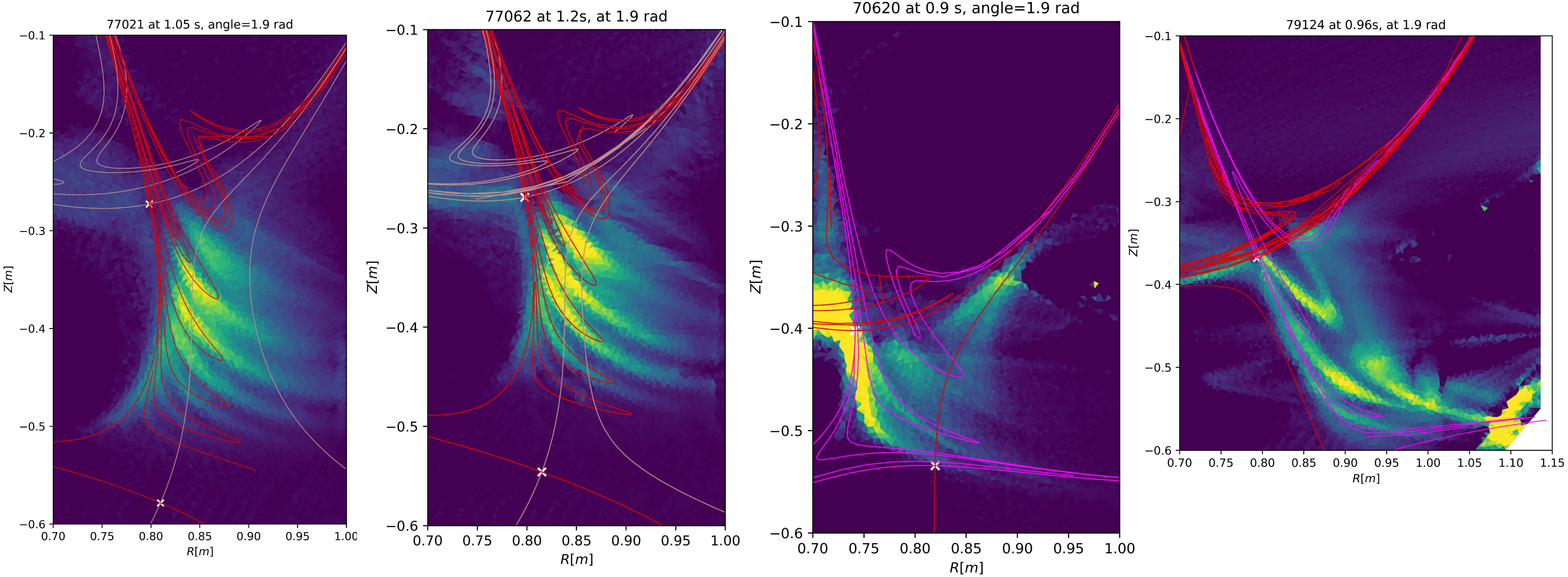



$$\Delta\Psi_s = \frac{\partial M_{sc}}{\partial X_c}\Delta X_c I_c \simeq \frac{\partial M_{sc}}{\partial R_c}\Delta R_c \cos\theta_s I_c$$

note: R derivative of elongation of coil not R center coordinate

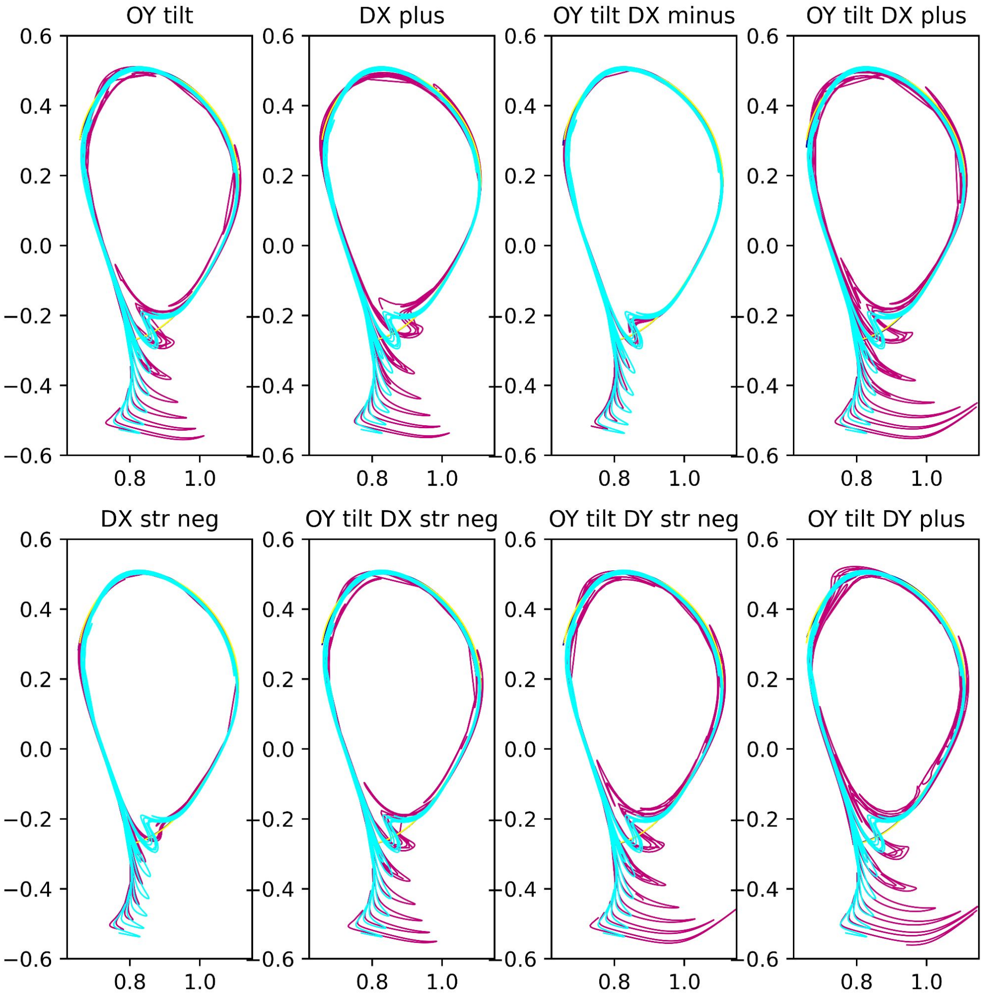


77021 at 1.05 s, angle=1.9 rad
77062 at 1.2s, at 1.9 rad
79124 at 0.96s, at 1.9 rad
70620 at 0.9 s, angle=1.9 rad
80064 at 1.1 s, angle=1.9 rad
Z[m]
R[m]
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
0.8
1.0




leg length / periodicity: inaccuracy in manifold calculation or other mechanism (transport, drift…)

- use symmetry to reduce degrees of freedom
- combine tilt, constant displacement and linear displacement
- elongation possible but not periodicity reduction



## to do:

- write my master’s thesis
- wait for the new MANTIS shot and get FastCam images
- add heteroclinic points and turnstile area
- make a 1-D line intercepting manifold/legs -> show periodicity doesn’t match